
Does Cost-Efficiency Travel? Accuracy and Cost Rank Transfer in Fixed-Scaffold Agent Evaluation

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Abstract

1 Agent leaderboards increasingly report both task success and dollar cost, inviting
2 practitioners to select a “cost-efficient” model from a single benchmark. We test
3 whether this selection signal transfers across workloads. Using public results from
4 the Holistic Agent Leaderboard (HAL), we hold the displayed agent scaffold fixed to
5 the HAL Generalist Agent and compare repeated displayed model labels across six
6 benchmarks. Accuracy-rank association is often positive across benchmark pairs,
7 and robust dollar-cost-rank transfer is likewise positive in several high-overlap
8 pairs: cost Spearman correlation ranges from 0.74 to 0.93 among pairs with at least
9 12 shared labels. Across all eligible pairs, cost point estimates range from -0.75 to
10 0.93 , but every negative estimate occurs at $N \leq 9$, has a configuration-resampling
11 interval crossing zero, and does not survive multiplicity correction or a label-
12 permutation null. Pareto-frontier membership remains non-universal: no broadly
13 tested label lies on every cost–accuracy frontier under three frontier definitions. We
14 also distinguish the ordinary nondominated frontier from a HAL-inspired convex-
15 hull reconstruction, finding that the public HAL labels align closely with the
16 latter but not identically. These descriptive results support a qualified conclusion:
17 some agent workloads exhibit robust positive cost-rank transfer, but sparse low-
18 overlap comparisons are too indeterminate to support universal efficiency claims or
19 confirmed workload-specific reversals. The study does not identify causal model or
20 scaffold effects: public display labels do not fully encode all benchmark-specific
21 setup details.

22 1 Introduction

23 Agent benchmarks now evaluate systems that reason over multiple steps, invoke tools, and interact
24 with environments. Unlike static model evaluations, an agent’s reported behavior depends on
25 an underlying language model, an agent scaffold, tools, prompts, execution limits, and the task
26 environment. These systems can also be expensive to evaluate: the Holistic Agent Leaderboard (HAL)
27 reports 21,730 rollouts across nine benchmarks and roughly \$40,000 in evaluation cost [Kapoor et al.,
28 2026]. Consequently, cost-aware leaderboards increasingly plot accuracy against dollar cost and
29 recommend points on a cost–accuracy frontier.

30 A practical inference often follows: if a model or agent is cost-efficient on one benchmark, it is a
31 promising low-cost choice elsewhere. This inference can hold if model capability and model cost are
32 portable; it can fail if token consumption, retry loops, tool calls, and stopping behavior change with
33 the workload. This paper asks a narrow descriptive question: *when the publicly displayed scaffold is*
34 *held fixed, how stable are repeated HAL displayed model labels’ accuracy ranks, dollar-cost ranks,*
35 *and cost–accuracy frontier positions across workloads?*

36 We conduct a secondary analysis of public HAL leaderboard data. The data are sparse and unbalanced
37 across scaffolds, so rather than attempt an unsupported causal decomposition of model, scaffold, and
38 workload effects, we focus on the only broadly repeated scaffold: the HAL Generalist Agent. Across
39 six benchmarks with sufficient overlap, we compare rankings only among shared displayed model
40 labels. Our main findings are:

- 41 • Accuracy and cost rank associations are robustly positive in several high-overlap pairs; lower-
42 overlap pairs are too uncertain to establish either transfer or reversal.
- 43 • Nondominated cost–accuracy frontier membership is non-universal: no broadly tested displayed
44 label lies on every available frontier under three decision-frontier definitions.
- 45 • Frontier conclusions depend on the decision model. The public HAL Is Pareto labels closely
46 match an origin-anchored convex-hull reconstruction, whereas standard nondominance retains
47 additional points.

48 The result is not universal non-transfer or universal transfer. Several high-overlap pairs show robust
49 positive association in both accuracy and dollar cost, while low-overlap apparent reversals are
50 statistically indeterminate under the checks in Section 6. Thus, transfer cannot be assumed for every
51 new workload from a single leaderboard.

52 2 Related work

53 **Cost-aware agent evaluation.** Kapoor et al. [2026] introduced HAL as a standardized, cost-aware
54 agent leaderboard spanning coding, web, science, and customer-service settings. HAL reports dollar-
55 and token-cost Pareto curves and documents substantial scaffold sensitivity. Kapoor et al. [2024]
56 argue that agent evaluation should jointly consider accuracy and cost, using Pareto plots to expose
57 avoidable expenditure. SWE-Effi studies resource-constrained software-engineering agents and
58 reports that effectiveness depends on scaffold–model integration [Fan et al., 2025]. These works
59 establish the importance of cost and scaffold effects, but do not systematically measure whether a
60 cost–accuracy ordering transfers between distinct workloads under a fixed scaffold.

61 **Stability of agent and benchmark rankings.** Ndzomga [2026] ask whether smaller task subsets
62 can preserve *within-benchmark* agent rankings under scaffold and temporal shift. Their target is
63 evaluation reduction, not cross-workload transfer of cost–accuracy trade-offs. In the static-LLM set-
64 ting, Benchmark² proposes cross-benchmark ranking consistency based on Kendall correlation [Qian
65 et al., 2026]. We adapt the cross-benchmark ranking perspective to interactive agents and separately
66 analyze accuracy ranks, dollar-cost ranks, and frontier-set overlap.

67 3 Data, scope, and coverage

68 We use the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026].
69 The frozen file contains 242 displayed evaluations across nine benchmarks, 11 displayed scaffolds,
70 and 29 displayed model labels. Accuracy and total dollar cost are complete, but 235 rows report a
71 single run and only seven report two runs. The data therefore support descriptive rank and frontier
72 analysis, not rollout-level uncertainty estimation.

73 Coverage determines the design. Most scaffolds occur on only one benchmark; the HAL Generalist
74 Agent occurs on seven. We retain the six Generalist benchmarks with adequate overlap: CORE-Bench
75 Hard (21 labels), GAIA (17), SciCode (9), ScienceAgentBench (7), SWE-bench Verified Mini (18),
76 and TAU-bench Airline (14). We exclude USACO because it contains one Generalist row. Figure 1
77 shows the resulting incomplete label-by-benchmark matrix.

78 The repeated unit is an exact public Models display string within the exact public Agent Name
79 display string. This includes the visible date and reasoning suffix when present. It does *not* fully
80 specify all benchmark-specific prompts, tools, budgets, or harness settings. Thus, the analysis
81 describes repeated public HAL labels under a repeated displayed scaffold; it is not a controlled
82 base-model experiment.

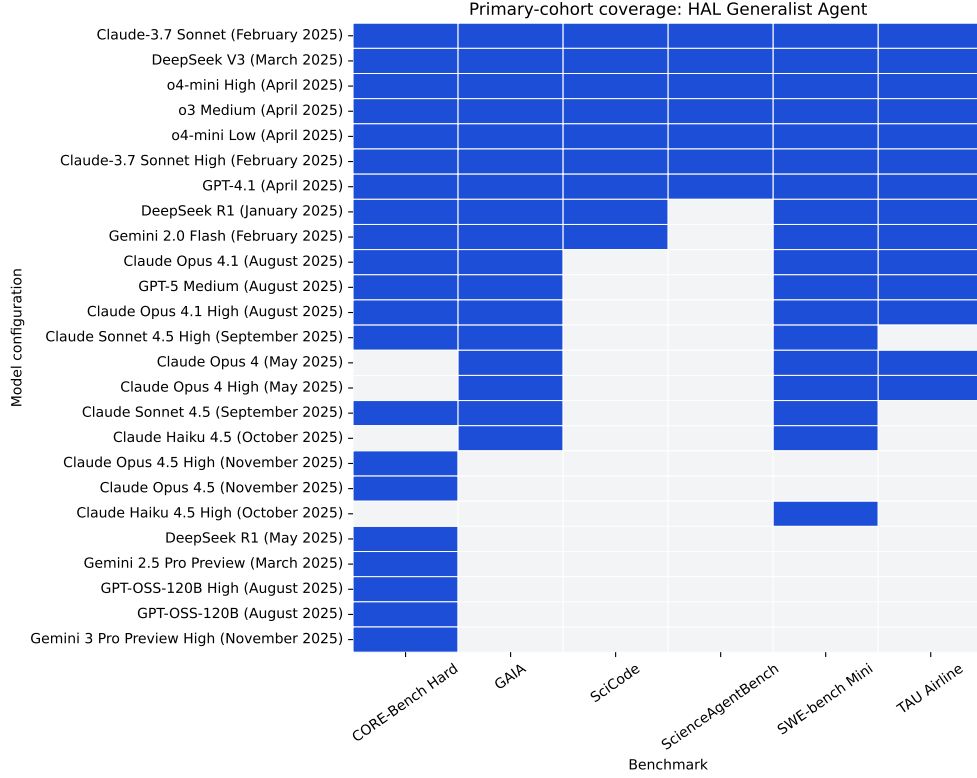


Figure 1: Coverage of the primary cohort. A filled cell indicates that the public HAL table contains the displayed model label under the displayed HAL Generalist Agent scaffold. Sparse coverage motivates pairwise analyses restricted to shared labels.

83 4 Method

84 **Rank transfer.** For each benchmark pair with at least five shared displayed labels, we compute
 85 Spearman rank correlation ρ and Kendall’s τ_b separately for accuracy and total dollar cost. Let L_{ij}
 86 be labels observed in both benchmarks i and j . We correlate their displayed accuracy values and,
 87 separately, their displayed total costs. We report $|L_{ij}|$ for every pair.

88 To characterize sensitivity to the finite label set, we use 5,000 percentile *configuration-resampling*
 89 draws: resample shared labels with replacement and recompute the correlation. These intervals are
 90 neither rollout-level uncertainty intervals nor population-model confidence intervals. Degenerate
 91 draws without rank variation are excluded and counted. We do not use interval inclusion of zero for
 92 hypothesis testing.

93 **Two frontier definitions.** For lower cost and higher accuracy, the nondominated frontier includes
 94 point x if no observed point y has cost no greater than x , accuracy at least that of x , and at least one
 95 strict improvement. This corresponds to choosing one displayed configuration. HAL’s public analysis
 96 code instead adds the origin and computes an upper cost–accuracy hull, which has a randomized-
 97 policy interpretation: a mixture of two systems can dominate an interior point in expectation. We
 98 implement a *HAL-inspired convex-hull reconstruction* based on the public reference code, while
 99 avoiding a claim of exact identity because two supplied labels differ in this frozen snapshot. For both
 100 definitions, we report pairwise Jaccard similarity of frontier sets. We distinguish full-cohort frontiers
 101 from frontiers recomputed only on a pair’s shared labels.

102 **Scaffold sensitivity.** As a secondary descriptive check, we compare shared displayed labels across
 103 scaffolds *within* a benchmark, reporting paired accuracy differences and log cost ratios. This confirms
 104 prior scaffold sensitivity rather than estimating a general scaffold effect.

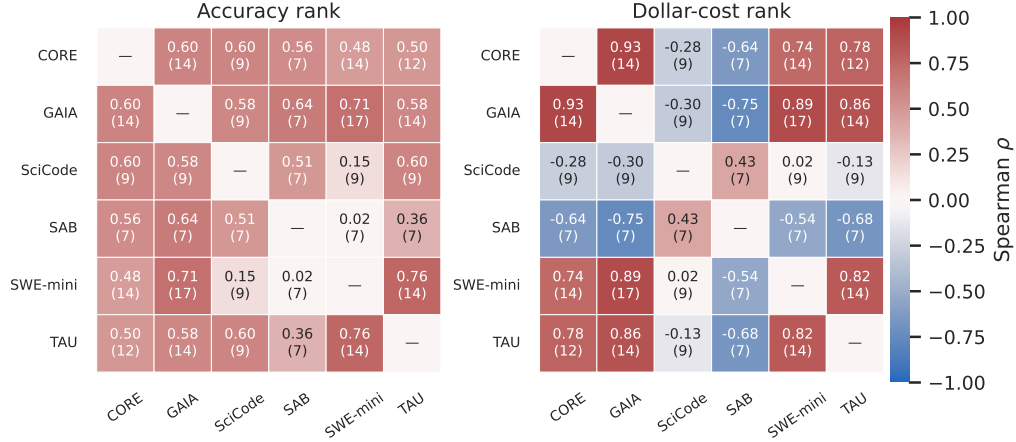


Figure 2: Cross-workload rank-transfer point estimates under the displayed HAL Generalist Agent scaffold. Each off-diagonal cell reports Spearman ρ and the number of shared displayed labels. Robustness checks show that several high-overlap positive associations survive multiplicity correction, whereas negative low-overlap point estimates are statistically indeterminate.

Table 1: Higher-overlap benchmark pairs. Brackets are 5,000-draw configuration-resampling sensitivity intervals, not rollout-level confidence intervals. ND denotes nondominated frontier membership recomputed on the shared-label cohort.

Pair	N	Accuracy ρ	Cost ρ	ND Jaccard
CORE-Bench Hard–GAIA	14	0.60 [0.16, 0.81]	0.93 [0.72, 0.99]	0.43
CORE-Bench Hard–SWE-bench Mini	14	0.48 [-0.05, 0.82]	0.74 [0.29, 0.93]	0.25
CORE-Bench Hard–TAU Airline	12	0.50 [-0.12, 0.86]	0.78 [0.26, 0.98]	0.17
GAIA–SWE-bench Mini	17	0.71 [0.31, 0.91]	0.89 [0.61, 0.99]	0.33
GAIA–TAU Airline	14	0.58 [0.14, 0.85]	0.86 [0.58, 0.97]	0.29
SWE-bench Mini–TAU Airline	14	0.76 [0.40, 0.91]	0.82 [0.46, 0.97]	0.43

5 Results

5.1 Rank transfer is robust in high-overlap pairs but indeterminate in sparse pairs

Figure 2 summarizes point estimates. Several high-overlap pairs exhibit positive association in both metrics: GAIA–SWE-bench Mini has 17 shared labels with $\rho = 0.71$ for accuracy and 0.89 for cost; SWE-bench Mini–TAU Airline has $\rho = 0.76$ and 0.82, respectively. The 10,000-draw configuration-resampling and permutation checks in Section 6 support these high-overlap positive associations.

Some low-overlap pairs have negative cost point estimates despite positive accuracy point estimates. For example, GAIA–ScienceAgentBench has accuracy $\rho = 0.64$ and cost $\rho = -0.75$ over seven shared labels. However, this cost interval is $[-1.00, 0.17]$, it does not survive Holm correction, and its permutation $p = 0.0675$. We therefore treat such cases as sparse-pair heterogeneity rather than confirmed cost-rank reversals. SciCode–SWE-bench Mini is near zero for both metrics ($\rho = 0.15$ accuracy, 0.02 cost, $N = 9$), with broad sensitivity intervals.

Table 1 lists the six pairs with at least 12 shared labels. These are the primary inferentially useful comparisons: cost ranks are strongly aligned and remain non-null under configuration resampling, permutation, and Holm correction. Nondominated frontier Jaccard similarity is nonetheless low to moderate (0.17–0.43), showing that binary frontier composition can change even where scalar ranks agree.

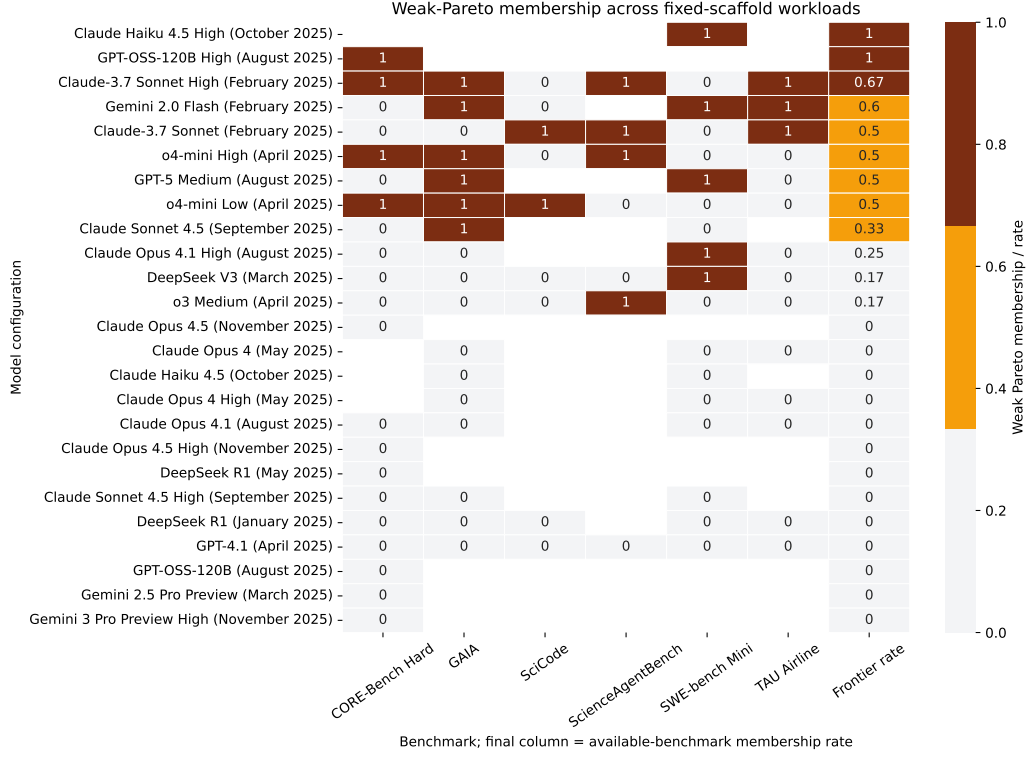


Figure 3: Nondominated cost–accuracy membership across primary workloads. The final column is each label’s rate over its available workloads. No label tested on at least five workloads is nondominated on all of them.

Table 2: Nondominated frontier appearances for labels tested on at least five primary workloads.

Model label	Tested	ND appearances	ND rate
Claude-3.7 Sonnet High	6	4	0.67
Claude-3.7 Sonnet	6	3	0.50
o4-mini High	6	3	0.50
o4-mini Low	6	3	0.50
DeepSeek V3	6	1	0.17
o3 Medium	6	1	0.17
GPT-4.1	6	0	0.00
Gemini 2.0 Flash	5	3	0.60
DeepSeek R1	5	0	0.00

5.2 Frontier membership is not portable

Figure 3 displays nondominated membership by workload. Among labels tested on at least five workloads, Claude-3.7 Sonnet High has the highest nondominated appearance count, four of six; no broadly tested label is nondominated everywhere. Table 2 summarizes the broadly covered labels. Because membership is discontinuous near a frontier, we interpret this result together with the continuous rank results rather than as a standalone universal ranking.

5.3 Frontier definition and scaffold comparisons matter

The source HAL labels recover 30 frontier rows under the hull reconstruction, with two discrepancies; standard nondominance includes 19 additional non-dominated rows. This is substantively relevant: a discrete buyer choosing one agent and a deployer willing to randomize between agents solve different decision problems.

134 Within individual benchmarks, scaffold changes are also large and direction-dependent. For example,
 135 across 15 shared labels on SWE-bench Mini, the displayed SWE-Agent is on average 27.4 percentage
 136 points more accurate than the displayed Generalist scaffold, with a higher median cost. Across seven
 137 shared labels on ScienceAgentBench, SAB Self-Debug is both more accurate on average and lower
 138 cost. These benchmark-specific contrasts are consistent with HAL and SWE-Effi, and motivate rather
 139 than replace the fixed-scaffold primary design.

140 6 Robustness checks and threats to validity

141 We added four checks aimed at likely reviewer concerns. First, Table 3 reports 10,000-draw
 142 configuration-resampling intervals for all 15 eligible pairs. The strong positive associations in
 143 high-overlap pairs remain stable: for example, cost ρ is 0.93 [0.72, 0.99] for CORE-GAIA and 0.89
 144 [0.60, 0.99] for GAIA-SWE-mini. In contrast, every negative cost estimate occurs at $N \leq 9$ and
 145 its interval crosses zero; GAIA-ScienceAgentBench is -0.75 [-1.00, 0.17]. These are sensitivity
 146 intervals over the observed labels, not rollout-level confidence intervals.

147 Second, we apply Holm-Bonferroni correction separately to 15 accuracy and 15 cost rank tests
 148 (Table 4). Two accuracy correlations survive: GAIA-SWE-mini and SWE-mini-TAU. Six positive
 149 cost correlations survive: CORE-GAIA, CORE-SWE-mini, CORE-TAU, GAIA-SWE-mini, GAIA-
 150 TAU, and SWE-mini-TAU. No negative cost correlation survives. A 10,000-draw label-permutation
 151 null gives the same message: the six high-overlap positive cost correlations are distinguishable from
 152 random pairing ($p \leq 0.0045$), whereas the negative low-overlap correlations are not.

153 Third, rank changes are concrete but pair-dependent. For GAIA-ScienceAgentBench, 1/7 shared la-
 154 bels moves from the cheapest quartile to the most expensive quartile, 3/7 make an extreme top/bottom
 155 flip in either direction, and 6/7 change cost quartile. For the high-overlap GAIA-SWE-mini pair, no
 156 extreme cost-quartile flip occurs and only 3/17 labels change quartile. Table 5 reports all cost-quartile
 157 flips.

158 Finally, the universal-winner conclusion depends on its exact wording. Table 6 shows that no label
 159 tested on at least five workloads is on every frontier under standard nondominance, the HAL-inspired
 160 hull, or a conservative 5% near-tie tolerance. However, the stronger statement that no label appears
 161 on more than 4/6 frontiers is not robust: under the tolerance sensitivity, o4-mini Low appears on 5/6.
 162 We therefore retain only the robust “no universal broad winner” conclusion.

Table 3: All pairwise Spearman correlations and 10,000-draw configuration-resampling sensitivity intervals. These intervals describe finite-label sensitivity, not rollout-level uncertainty.

Pair	N	Acc. ρ [interval]	Cost ρ [interval]
CORE-Bench Hard-GAIA	14	0.60 [0.16, 0.81]	0.93 [0.72, 0.99]
CORE-Bench Hard-SciCode	9	0.60 [-0.24, 0.92]	-0.28 [-0.86, 0.51]
CORE-Bench Hard-ScienceAgentBench	7	0.56 [-0.41, 1.00]	-0.64 [-1.00, 0.17]
CORE-Bench Hard-SWE-bench Mini	14	0.48 [-0.04, 0.82]	0.74 [0.27, 0.93]
CORE-Bench Hard-TAU Airline	12	0.50 [-0.11, 0.85]	0.78 [0.26, 0.98]
GAIA-SciCode	9	0.58 [-0.19, 0.98]	-0.30 [-0.89, 0.55]
GAIA-ScienceAgentBench	7	0.64 [-0.18, 1.00]	-0.75 [-1.00, 0.17]
GAIA-SWE-bench Mini	17	0.71 [0.32, 0.91]	0.89 [0.60, 0.99]
GAIA-TAU Airline	14	0.58 [0.16, 0.85]	0.86 [0.57, 0.97]
SciCode-ScienceAgentBench	7	0.51 [-0.64, 1.00]	0.43 [-0.62, 1.00]
SciCode-SWE-bench Mini	9	0.15 [-0.66, 0.78]	0.02 [-0.81, 0.81]
SciCode-TAU Airline	9	0.60 [0.00, 0.92]	-0.13 [-0.80, 0.76]
ScienceAgentBench-SWE-bench Mini	7	0.02 [-0.97, 0.65]	-0.54 [-1.00, 0.65]
ScienceAgentBench-TAU Airline	7	0.36 [-0.76, 0.96]	-0.68 [-1.00, 0.19]
SWE-bench Mini-TAU Airline	14	0.76 [0.41, 0.91]	0.82 [0.46, 0.97]

Table 4: Uncorrected and Holm-Bonferroni-adjusted rank-correlation tests, corrected separately within accuracy and cost families (15 tests each).

Pair	Metric	Uncorrected p	Holm p	Reject
corebench_hard-gaia	Accuracy	0.0222	0.2891	No
corebench_hard-scicode	Accuracy	0.0903	0.9328	No
corebench_hard-scienceagentbench	Accuracy	0.1925	0.9623	No
corebench_hard-swebench_verified_mini	Accuracy	0.0848	0.9328	No
corebench_hard-taubench_airline	Accuracy	0.0965	0.9328	No
gaia-scicode	Accuracy	0.1012	0.9328	No
gaia-scienceagentbench	Accuracy	0.1194	0.9328	No
gaia-swebench_verified_mini	Accuracy	0.0015	0.0222	Yes
gaia-taubench_airline	Accuracy	0.0294	0.3524	No
scicode-scienceagentbench	Accuracy	0.2474	0.9898	No
scicode-swebench_verified_mini	Accuracy	0.6992	1.0000	No
scicode-taubench_airline	Accuracy	0.0886	0.9328	No
scienceagentbench-swebench_verified_mini	Accuracy	0.9694	1.0000	No
scienceagentbench-taubench_airline	Accuracy	0.4271	1.0000	No
swebench_verified_mini-taubench_airline	Accuracy	0.0017	0.0238	Yes
corebench_hard-gaia	Cost	0.0000	0.0000	Yes
corebench_hard-scicode	Cost	0.4600	1.0000	No
corebench_hard-scienceagentbench	Cost	0.1194	0.8357	No
corebench_hard-swebench_verified_mini	Cost	0.0027	0.0294	Yes
corebench_hard-taubench_airline	Cost	0.0030	0.0299	Yes
gaia-scicode	Cost	0.4328	1.0000	No
gaia-scienceagentbench	Cost	0.0522	0.4696	No
gaia-swebench_verified_mini	Cost	0.0000	0.0000	Yes
gaia-taubench_airline	Cost	0.0001	0.0009	Yes
scicode-scienceagentbench	Cost	0.3374	1.0000	No
scicode-swebench_verified_mini	Cost	0.9661	1.0000	No
scicode-taubench_airline	Cost	0.7324	1.0000	No
scienceagentbench-swebench_verified_mini	Cost	0.2152	1.0000	No
scienceagentbench-taubench_airline	Cost	0.0938	0.7500	No
swebench_verified_mini-taubench_airline	Cost	0.0003	0.0035	Yes

163 A Robustness checks

164 **Threats to validity.** Our conclusions are subject to construct, internal, and external validity threats.
165 Construct validity is limited because the public Models and Agent Name display strings do not
166 fully encode benchmark-specific prompts, tools, budgets, harness versions, or pricing assumptions;
167 matching a displayed label under the displayed Generalist scaffold is therefore not a controlled base-
168 model comparison. Internal validity is limited by sparse, unbalanced coverage and primarily single-
169 run evaluations. Two broadly covered labels are absent from ScienceAgentBench, but the public
170 table provides no failure code or exclusion rationale; this prevents us from assuming missingness at
171 random or attributing exclusions to capability or cost. Configuration-resampling intervals quantify
172 sensitivity to the observed shared-label set rather than rollout variance, and all negative dollar-cost
173 correlations occur in low-overlap pairs whose intervals cross zero and whose permutation tests do
174 not reject random pairing. Finally, external validity is restricted to this frozen HAL snapshot, its
175 then-current dollar prices, and six workloads under one repeated displayed scaffold. We therefore

Table 5: Cost-quartile flips within shared-label benchmark pairs. “Cheap-to-expensive” means top-quartile cheapest on the first benchmark and bottom-quartile most expensive on the second.

Pair	N	Cheap-to-expensive	Any extreme flip
corebench_hard-gaia	14	0/14 (0%)	0/14 (0%)
corebench_hard-scicode	9	1/9 (11%)	2/9 (22%)
corebench_hard-scienceagentbench	7	1/7 (14%)	2/7 (29%)
corebench_hard-swebench_verified_mini	14	1/14 (7%)	1/14 (7%)
corebench_hard-taubench_airline	12	0/12 (0%)	0/12 (0%)
gaia-scicode	9	1/9 (11%)	3/9 (33%)
gaia-scienceagentbench	7	1/7 (14%)	3/7 (43%)
gaia-swebench_verified_mini	17	0/17 (0%)	0/17 (0%)
gaia-taubench_airline	14	0/14 (0%)	0/14 (0%)
scicode-scienceagentbench	7	0/7 (0%)	0/7 (0%)
scicode-swebench_verified_mini	9	2/9 (22%)	3/9 (33%)
scicode-taubench_airline	9	2/9 (22%)	3/9 (33%)
scienceagentbench-swebench_verified_mini	7	2/7 (29%)	3/7 (43%)
scienceagentbench-taubench_airline	7	2/7 (29%)	2/7 (29%)
swebench_verified_mini-taubench_airline	14	0/14 (0%)	0/14 (0%)

Table 6: Sensitivity of the universal-winner claim to the frontier definition. Broad labels are tested on at least five workloads.

Definition	Max appearances (broad labels)	Universal broad winner
Nondominated	4	No
HAL-inspired hull	3	No
Nondominated, 5% tolerance	5	No

interpret robust positive high-overlap correlations as descriptive evidence of transfer in selected workload pairs, and interpret low-overlap reversals as unresolved heterogeneity that motivates rather than proves workload-specific cost dynamics.

B Discussion

The practical implication is conditional, not absolute. A cost-efficient point on one benchmark can be a useful prior, and several high-overlap benchmark pairs show robust positive cost-rank transfer. However, sparse low-overlap pairs provide insufficient evidence to extrapolate that relationship universally or to establish true reversals. Dollar cost plausibly responds to workload-specific execution dynamics—tool calls, retries, loop length, and termination behavior—in addition to model pricing. A practitioner selecting an agent for a new workload should therefore validate cost on representative tasks rather than extrapolate from a single efficiency leaderboard.

The analysis also highlights a reporting issue. A frontier label is not self-explanatory: ordinary non-dominance and a convex-hull/randomization frontier encode different deployment choices. Leaderboards should state which decision model they use and, when feasible, expose both.

C Limitations

This is a secondary analysis of a sparse public snapshot. First, repeated coverage is incomplete; most scaffolds occur on only one benchmark, and the primary fixed-scaffold analysis retains only six workloads. Second, public display labels do not fully encode benchmark-specific prompts, tool policies, budgets, or harness versions. Holding a display label fixed does not isolate a causal base-model effect. Third, most evaluations are single-run, so configuration-resampling intervals do not measure rollout variance. Fourth, dollar cost combines observed execution with pricing assumptions; the supplied CSV lacks token totals, preventing a token-cost robustness analysis. Fifth, only three same-domain pairs are available under HAL’s categories, so domain conclusions are exploratory and omitted from the main claim. Finally, frontier membership is sensitive to the comparator set and small changes near the frontier; we therefore report both full-cohort and shared-cohort analyses in the supplemental material.

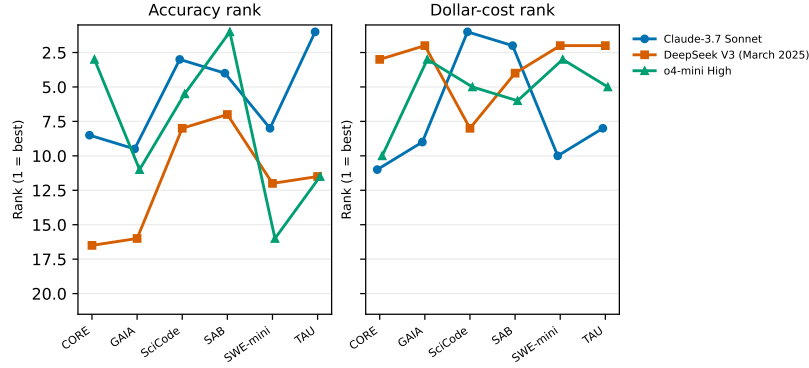


Figure 4: Supplementary rank-line view for three labels observed on all six primary workloads. Ranks are among all displayed Generalist labels available on each benchmark; lower is better. Ties use average ranks, and markers are slightly offset horizontally only to keep tied series visible. The panel illustrates that rank movement differs by metric and workload; it is not used for inference.

D Conclusion

Across public HAL results with a repeated displayed Generalist scaffold, several high-overlap workload pairs show robust positive accuracy- and dollar-cost-rank transfer, while sparse low-overlap comparisons are too uncertain to establish either transfer or reversal. Cost-accuracy frontier membership is non-universal under multiple decision-frontier definitions. Thus, a single benchmark can supply a useful prior but not a universal efficiency guarantee. Future agent leaderboards should use crossed model-scaffold-benchmark designs with repeated runs and report both token- and dollar-based decision frontiers.

References

- Zhiyu Fan, Kirill Vasilevski, Dayi Lin, et al. SWE-effi: Re-evaluating software AI agent system effectiveness under resource constraints. *arXiv preprint arXiv:2509.09853*, 2025. URL <https://arxiv.org/abs/2509.09853>.
- Sayash Kapoor et al. AI agents that matter. *arXiv preprint arXiv:2407.01502*, 2024. URL <https://arxiv.org/abs/2407.01502>.
- Sayash Kapoor et al. Holistic agent leaderboard: The missing infrastructure for AI agent evaluation. *arXiv preprint arXiv:2510.11977*, 2026. URL <https://arxiv.org/abs/2510.11977>.
- Franck Ndzomga. Efficient benchmarking of AI agents. *arXiv preprint arXiv:2603.23749*, 2026. URL <https://arxiv.org/abs/2603.23749>.
- Qi Qian, Chengsong Huang, et al. Benchmark²: Systematic evaluation of LLM benchmarks. *arXiv preprint arXiv:2601.03986*, 2026. URL <https://arxiv.org/abs/2601.03986>.

222 E Frontier-label reproducibility audit

Table 7: Supplied HAL frontier labels compared with reconstructed frontiers by benchmark. Agreement is the fraction of displayed rows with identical Boolean labels.

Benchmark	Rows	Supplied	Nondominated	Hull	Supplied-ND agreement	Supplied-hull agreement
assistantbench	15	2	3	2	0.933	1.000
corebench_hard	45	5	10	5	0.889	1.000
gaia	32	4	6	4	0.938	1.000
online_mind2web	22	3	4	3	0.955	1.000
scicode	33	2	3	3	0.970	0.970
scienceagentbench	23	4	8	4	0.826	1.000
swebench_verified_mini	33	3	7	3	0.879	1.000
taubench_airline	26	3	3	4	1.000	0.962
usaco	13	4	5	4	0.923	1.000

223 F Reproducibility details

224 The analysis freezes the public CSV using the recorded source hash in the anonymous supplement.
 225 It uses exact public display labels, minimum pairwise overlap five, 5,000 configuration-resampling
 226 draws, and random seed 20260816. The released anonymous supplement should include the analysis
 227 code, input provenance manifest, generated tables, and commands needed to regenerate all figures.

228 NeurIPS Paper Checklist

229 1. Claims

230 Question: Do the main claims made in the abstract and introduction accurately reflect the
 231 paper’s contributions and scope?

232 Answer: [\[Yes\]](#)

233 Justification: The abstract and introduction state that the analysis is descriptive, restricted
 234 to repeated public HAL display labels under a repeated displayed scaffold, and does not
 235 identify causal model or scaffold effects.

236 2. Limitations

237 Question: Does the paper discuss the limitations of the work performed by the authors?

238 Answer: [\[Yes\]](#)

239 Justification: Section 6 discusses sparse coverage, display-label ambiguity, single-run
 240 evaluations, price dependence, missing token totals, limited domain comparisons, and
 241 frontier sensitivity.

242 3. Theory assumptions and proofs

243 Question: For each theoretical result, does the paper provide the full set of assumptions and
 244 a complete (and correct) proof?

245 Answer: [\[N/A\]](#)

246 Justification: The paper presents an empirical secondary analysis and no theoretical results.

247 4. Experimental result reproducibility

248 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
 249 perimental results of the paper to the extent that it affects the main claims and/or conclusions
 250 of the paper (regardless of whether the code and data are provided or not)?

251 Answer: [\[Yes\]](#)

252 Justification: Sections 3–4 and the appendix specify the public source, frozen hash, inclusion
 253 criteria, matching unit, frontier definitions, overlap threshold, resampling procedure, and
 254 seed.

255 5. Open access to data and code

256 Question: Does the paper provide open access to the data and code, with sufficient instruc-
 257 tions to faithfully reproduce the main experimental results, as described in supplemental
 258 material?

259 Answer: [Yes]

260 Justification: An anonymized supplement will contain the analysis code, provenance mani-
 261 fest, and reproduction instructions; the underlying public data source is cited.

262 **6. Experimental setting/details**

263 Question: Does the paper specify all the training and test details (e.g., data splits, hyperpa-
 264 rameters, how they were chosen, type of optimizer) necessary to understand the results?

265 Answer: [Yes]

266 Justification: No models are trained. The data filtering, rank metrics, frontier definitions,
 267 resampling count, and seed are specified in Sections 3–4.

268 **7. Experiment statistical significance**

269 Question: Does the paper report error bars suitably and correctly defined or other appropriate
 270 information about the statistical significance of the experiments?

271 Answer: [Yes]

272 Justification: We report configuration-resampling sensitivity intervals and explicitly state
 273 that they are not rollout-level or population-model confidence intervals and are not used for
 274 hypothesis tests.

275 **8. Experiments compute resources**

276 Question: For each experiment, does the paper provide sufficient information on the com-
 277 puter resources (type of compute workers, memory, time of execution) needed to reproduce
 278 the experiments?

279 Answer: [N/A]

280 Justification: The work is a lightweight secondary analysis of public tabular data and uses
 281 no model training or agent rollouts.

282 **9. Code of ethics**

283 Question: Does the research conducted in the paper conform, in every respect, with the
 284 NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

285 Answer: [Yes]

286 Justification: The work uses public aggregate evaluation data, attributes source authors,
 287 reports limitations, and makes no claims about people or protected groups.

288 **10. Broader impacts**

289 Question: Does the paper discuss both potential positive societal impacts and negative
 290 societal impacts of the work performed?

291 Answer: [N/A]

292 Justification: The work is a methodological analysis of public leaderboard measurements.
 293 Its practical implication is limited to more cautious agent selection and evaluation reporting.

294 **11. Safeguards**

295 Question: Does the paper describe safeguards that have been put in place for responsible
 296 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
 297 image generators, or scraped datasets)?

298 Answer: [N/A]

299 Justification: The paper releases no trained model, agent capability, or new high-risk dataset.

300 **12. Licenses for existing assets**

301 Question: Are the creators or original owners of assets (e.g., code, data, models), used in
 302 the paper, properly credited and are the license and terms of use explicitly mentioned and
 303 properly respected?

304 Answer: [Yes]

305 Justification: The data and methodological sources are cited; the supplement records the
 306 upstream repository, commit, source path, and declared MIT license.

307 **13. New assets**

308 Question: Are new assets introduced in the paper well documented and is the documentation
 309 provided alongside the assets?

310 Answer: [Yes]

311 Justification: The analysis code and generated tables are documented by a source manifest,
 312 artifact manifest, tests, and reproduction commands.

313 **14. Crowdsourcing and research with human subjects**

314 Question: For crowdsourcing experiments and research with human subjects, does the paper
 315 include the full text of instructions given to participants and screenshots, if applicable, as
 316 well as details about compensation (if any)?

317 Answer: [N/A]

318 Justification: The study involves no crowdsourcing and no human subjects research.

319 **15. Institutional review board (IRB) approvals or equivalent for research with human**
 320 **subjects**

321 Question: Does the paper describe potential risks incurred by study participants, whether
 322 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
 323 approvals (or an equivalent approval/review based on the requirements of your country or
 324 institution) were obtained?

325 Answer: [N/A]

326 Justification: The study involves no human subjects research.

327 **16. Declaration of LLM usage**

328 Question: Does the paper describe the usage of LLMs if it is an important, original, or
 329 non-standard component of the core methods in this research?

330 Answer: [N/A]

331 Justification: The core method is deterministic analysis of public tabular data; no LLM is
 332 used as a component of the empirical method.